# Set Operations

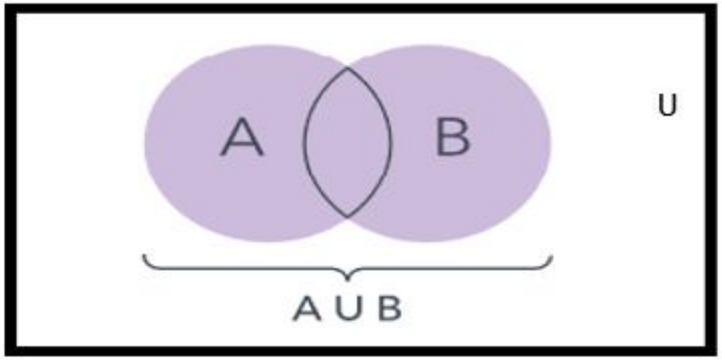
# Union

- **Definition**: Let $A$ and $B$ be sets. The *union* of the sets $A$ and $B$, denoted by $\boldsymbol{A} \cup \boldsymbol{B}$, is the set:

$$\{x | x \in A \vee x \in B\}$$

- **Example**: What is $\{1,2,3\} \cup \{3, 4, 5\}$?

**Solution**: $\{1,2,3,4,5\}$




All items in the sets.

**UNION REMARK:**

1. $A \cup B = B \cup A$ that is union is commutative you can prove this very easily only by using definition.

2. $A \subseteq A \cup B$ and $B \subseteq A \cup B$

The above remark of subset is easily seen by the definition of union.

# Intersection

- **Definition**: The *intersection* of sets $A$ and $B$, denoted by $\boldsymbol{A \cap B}$**,** is

$$\{x | x \in A \wedge x \in B\}$$

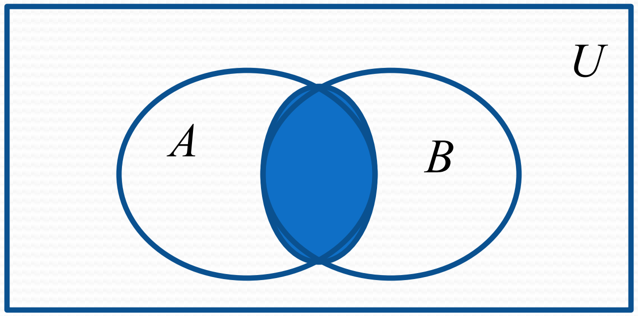
- Note if the intersection is empty, then $A$ and $B$ are said to be *disjoint*.
- **Example**: What is? $\{1,2,3\} \cap \{3,4,5\}$ ?

  **Solution**: $\{3\}$
- **Example:** What is?

  $\{1,2,3\} \cap \{4,5,6\}$ ?

  **Solution**: $\varnothing$

Venn Diagram for $A \cap B$



## INTERSECTION REMARKS:

1. $A \cap B = B \cap A$
2. $A \cap B \subseteq A$ and $A \cap B \subseteq B$
3. If $A \cap B = \varphi$, then A & B are called disjoint sets.

# Complement

**Definition**: If $A$ is a set, then the complement of the $A$ (with respect to $U$), denoted by **$\bar{A}$ is the set $U$ - $A$**

$$\bar{A} = \{x \in U \mid x \notin A\}$$

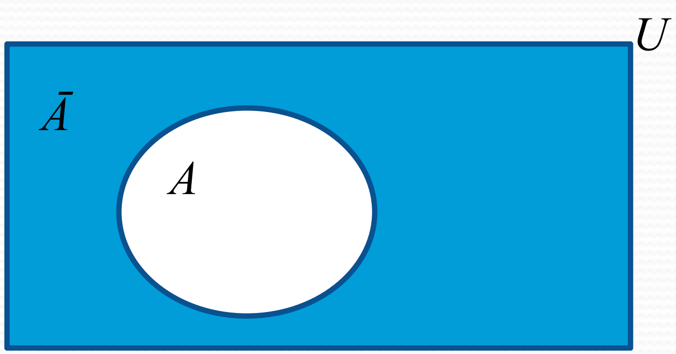
(The complement of A is sometimes denoted by $A^c$.)

**Example**: If $U$ is the positive integers less than 100, what is the complement of $\{x \mid x > 70\}$

Solution: $\{x \mid x \leq 70\}$

- **Example 2**:
- U = {a, b, c, d, e, f, g],
- A = {a, c, e, g}
- Then **$A^c$ = {b, d, f}**

Venn Diagram for Complement



## COMPLEMENT REMARKS:

1. $A^c = U - A$
2. $A \cap A^c = \varphi$
3. $A \cup A^c = U$

# Difference

- **Definition**: Let $A$ and $B$ be sets. The *difference* of $A$ and $B$, denoted by $A - B$, is the set containing the elements of $A$ that are not in $B$. The difference of $A$ and $B$ is also called the complement of $B$ with respect to $A$.

$$A - B = \{x \mid x \in \mathrm{A} \wedge x \notin B\} = A \cap \square B$$

**EXAMPLE:**

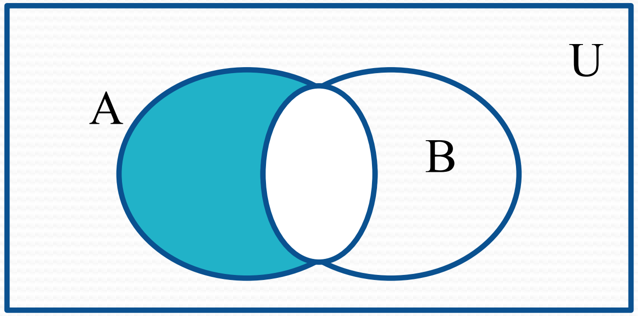
**U** = {a, b, c, d, e, f, g}

**A** = {a, c, e, g},

**B** = {d, e, f, g}

Then , **A – B = {a, c}**

Venn Diagram for $A - B$



## DIFFERENCE REMARKS

1. $A - B \neq B - A$ that is set difference is not commutative.
2. $A - B \subseteq A$

# Review Questions

**Example**: $U = \{0,1,2,3,4,5,6,7,8,9,10\}$ $A = \{1,2,3,4,5\}$, $B = \{4,5,6,7,8\}$

1. $A \cup B$

   **Solution:** $\{1,2,3,4,5,6,7,8\}$

2. $A \cap B$

   **Solution:** $\{4,5\}$

3. $\bar{A}$

   **Solution:** $\{0,6,7,8,9,10\}$

4. $\bar{B}$

   **Solution:** $\{0,1,2,3,9,10\}$

5. $A - B$

   **Solution:** $\{1,2,3\}$

6. $B - A$

   **Solution:** $\{6,7,8\}$

# Symmetric Difference (*optional*)

**Definition**: The *symmetric difference* of **A** and **B**, denoted by $A \oplus B$ is the set

$$(A - B) \cup (B - A)$$

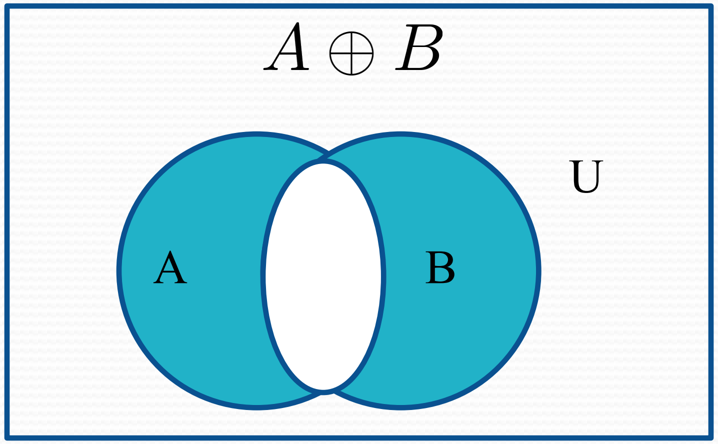
**Example**:

$U = \{0,1,2,3,4,5,6,7,8,9,10\}$

$A = \{1,2,3,4,5\}$ $B = \{4,5,6,7,8\}$

What is $A \oplus B$

- **Solution**: $\{1,2,3,6,7,8\}$




Venn Diagram

# SET IDENTITIES

| TABLE 1 Set Identities. | |
|---|---|
| ***Identity*** | ***Name*** |
| $A \cap U = A$<br>$A \cup \emptyset = A$ | Identity laws |
| $A \cup U = U$<br>$A \cap \emptyset = \emptyset$ | Domination laws |
| $A \cup A = A$<br>$A \cap A = A$ | Idempotent laws |
| $\overline{(\overline{A})} = A$ | Complementation law |
| $A \cup B = B \cup A$<br>$A \cap B = B \cap A$ | Commutative laws |
| $A \cup (B \cup C) = (A \cup B) \cup C$<br>$A \cap (B \cap C) = (A \cap B) \cap C$ | Associative laws |
| $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$<br>$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ | Distributive laws |
| $\overline{A \cap B} = \overline{A} \cup \overline{B}$<br>$\overline{A \cup B} = \overline{A} \cap \overline{B}$ | De Morgan's laws |
| $A \cup (A \cap B) = A$<br>$A \cap (A \cup B) = A$ | Absorption laws |
| $A \cup \overline{A} = U$<br>$A \cap \overline{A} = \emptyset$ | Complement laws |

# Proof De Morgan Law

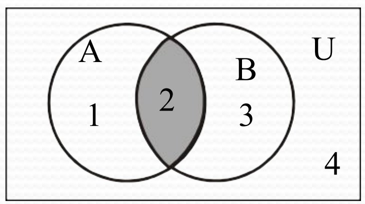
**Let A = {1,2} , B = {2,3} and U = {1,2,3,4}**

Prove the following using Venn Diagrams:

1. **$(A \cap B)^c = A^c \cup B^c$**

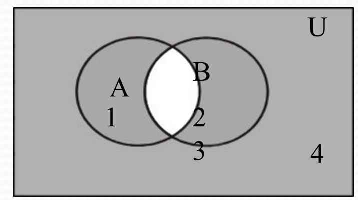
**SOLUTION(Proof using Venn Diagram):** Taking the L.H.S

(a)



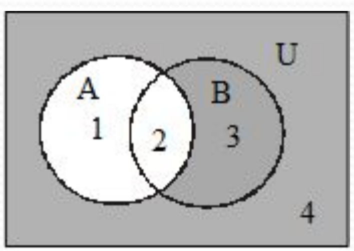

**$A \cap B$**

(b)

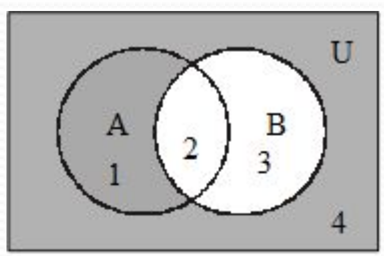



**$(A \cap B)^c$**

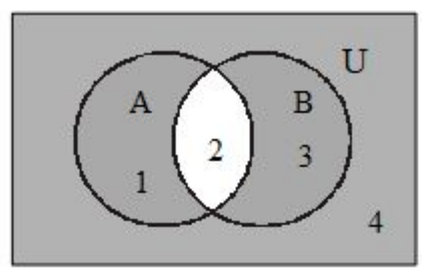
Now taking the R.H.S




$A^c$ is shaded.




$B^c$ is shaded.




$A^c \cup B^c$ is shaded.

**L.H.S = R.H.S , Hence Proved**

# TASK A

A.

**Let $U = \{1, 2, 3, \ldots, 10\}$,** **$X = \{1, 2, 3, 4, 5\}$**

**$Y = \{y \mid y = 2x, x \in X\}$,** **$Z = \{z \mid z^2 - 9z + 14 = 0\}$**

**Enumerate**
**:**

(1) $X \cap Y$ (2) $Y \cup Z$ (3) $X - Z$

(4) $Y^c$ (5) $X^c - Z^c$ (6) $(X - Z)^c$

# SOLUTION A

$U = \{1, 2, 3, \ldots, 10\}$,

$X = \{1, 2, 3, 4, 5\}$

$Y = \{y \mid y = 2x, x \in X\} = \{2, 4, 6, 8, 10\}$

$Z = \{z \mid z^2 - 9z + 14 = 0\} = \{2, 7\}$

(1) $X \cap Y = \{1, 2, 3, 4, 5\} \cap \{2, 4, 6, 8, 10\}$
$= \{2, 4\}$

(2) $Y \cup Z = \{2, 4, 6, 8, 10\} \cup \{2, 7\}$
$= \{2, 4, 6, 7, 8, 10\}$

(3) $X - Z = \{1, 2, 3, 4, 5\} - \{2, 7\}$
$= \{1, 3, 4, 5\}$

(4) $Y^c = U - Y = \{1, 2, 3, \ldots, 10\} - \{2, 4, 6, 8, 10\}$
$= \{1, 3, 5, 7, 9$

(5) $X^c - Z^c = \{6, 7, 8, 9, 10\} - \{1, 3, 4, 5, 6, 8, 9, 10\}$
$= \{7\}$

(6) $(X - Z)^c = U - (X - Z)$
$= \{1, 2, 3, \ldots, 10\} - \{1, 3, 4, 5\}$
$= \{2, 6, 7, 8, 9, 10\}$

**NOTE** $(X - Z)^c \neq X^c - Z^c$

# TASK B

2. **Given the following universal set U and its two subsets P and Q, where**

   $U = \{x \mid x \in Z, 0 \leq x \leq 10\}$

   $P = \{x \mid x \text{ is a prime number}\}$

   $Q = \{x \mid x^2 < 70\}$

**Draw a Venn diagram for the above and List the elements in $P^c \cap Q$.**

# SOLUTION B

First we write the sets in Tabular form.

$U = \{x \mid x \in Z, 0 \leq x \leq 10\}$

Since it is the set of integers that are greater then or equal 0 and less or equal to 10. So we have

$U = \{0, 1, 2, 3, \ldots, 10\}$

$P = \{x \mid x \text{ is a prime number}\}$

It is the set of prime numbers between 0 and 10. Remember Prime numbers are those numbers which have only two distinct divisors.

$P = \{2, 3, 5, 7\}$

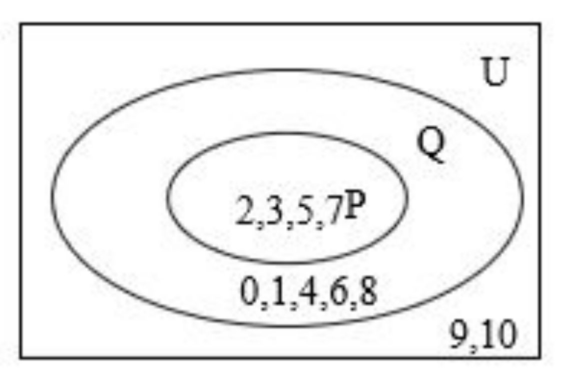
$Q = \{x \mid x^2 < 70\}$

The set Q contains the elements between 0 and 10 which has their square less or equal to 70.

Q= {0, 1, 2, 3, 4, 5

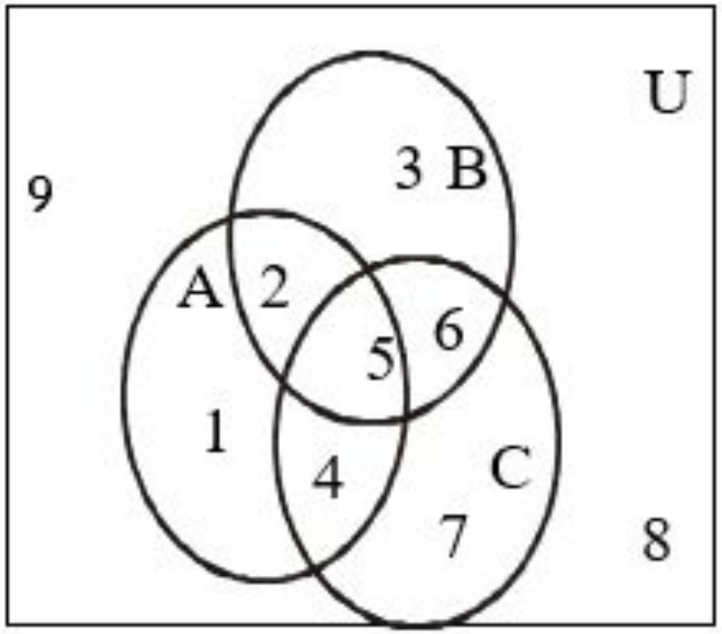
Thus we write the sets in

**<u>VENN DIAGRAM:</u>**



# TASK C

- **Use a Venn diagram to represent the following:**
  1. $(A \cap B) \cap C^c$
  2. $A^c \cup (B \cup C)$
  3. $(A - B) \cap C$
  4. $(A \cap B^c) \cup C^c$



# SOLUTION D

i. **$(A \cap B) \cap C^c$**

Let us first construct the sets from the given venn diagram.

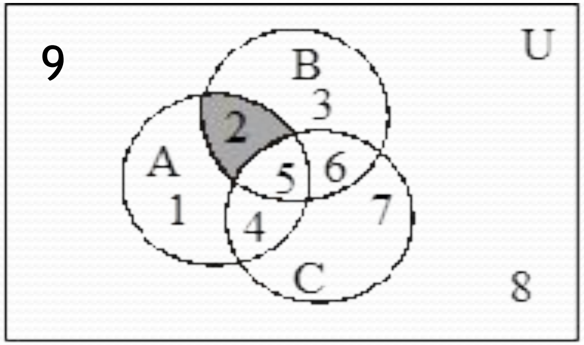
**A = { 1,2,4,5}   B = {2,3,5,6}   C = {4,5,6,7}   U = {1,2,3,4,5,6,7,8,9}**

NOW,

A ∩ **B = {2,5} and $C^c$ = U- C = {1,2,8,9}**

**SO,**

**$(A \cap B) \cap C^c$ = {2,5} ∩ {1,2,8,9} = {2}**




$(A \cap B) \cap C^c$ is shaded